

Mathematical Analysis Lecture 23. 积分学.

1. 可化为有理函数积分的例子: e.g. $\int R(x, \sqrt[n]{ax+b}) dx$ 令 $t = \sqrt[n]{ax+b}$

e.g. $\int R(x, \sqrt[n]{\frac{ax+b}{cx+d}}) dx$ 令 $t = \sqrt[n]{\frac{ax+b}{cx+d}}$

e.g. $\int R(x, \sqrt[m]{ax+b}, \sqrt[n]{ax+b}) dx$ 令 $t = \sqrt[p]{ax+b}$, p 为 m, n 的最小公倍数

例1: $\int \frac{dx}{1+\sqrt{x+2}}$ 令 $u = \sqrt[3]{x+2} \Rightarrow x = u^3 - 2$, 代回: 原式 $= 3 \int (u-1) + \frac{1}{1+u} du$.

例2: $\int \frac{1}{x} \cdot \sqrt{\frac{x+1}{x}} dx$ 令 $\sqrt{\frac{x+1}{x}} = t \Rightarrow x = \frac{1}{t^2-1}$, 原式 $= -2 \int \frac{t^2}{t^2-1} dt$.

例3: $\int \frac{dx}{\sqrt{x+3}\sqrt{x}}$ 令 $x = t^6$ 原 $= \int \frac{6t^5}{t^3+1} dt$

e.g. $\int R(\sin x, \cos x) dx$ 令 $t = \tan \frac{x}{2}$. $\sin x = \frac{2t \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$ $dx = \frac{2}{1+t^2} dt$.

例4: $\int \frac{1+\sin x}{3+\cos x} dx = \int \frac{1}{3+\cos x} dx - \int \frac{1}{3+\cos x} d(3+\cos x) = \dots$

例5: $\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} = \frac{1}{a^2} \int \frac{d \tan x}{\tan^2 x + \frac{b^2}{a^2}}$

例6: $\int \frac{1}{(a \sin x + b \cos x)^2} dx = \dots$ 令 $\frac{a}{\sqrt{a^2+b^2}} = \sin \varphi$, $\frac{b}{\sqrt{a^2+b^2}} = \cos \varphi$.

例7: $\int \frac{\cos^3 x - 2 \cos x}{1 + \sin^2 x + \sin^4 x} dx = - \int \frac{8 \sin^2 x + 1}{1 + \sin^2 x + \sin^4 x} d \sin x$